SESE3133: Formal Methods in Software Engineering

Spring 2026

Assignment 03

Total Marks: 40 **Obtained Marks: _____** **Date: 5/31/2026** **Section: SE_F3**

Name 1 : Muhammed Arham **ID1: L1_F23_BSSE_0372**

Name 2 : Muzaffar Ali **ID2: L1_F23_BSSE_0395**

CLO #	Keywords	Taxonomy Level
CLO 2	Built, Construct, Design	C3

Questions Mapped	Total Marks	Marks Obtained
1	10	
2	10	
3	15	
4	05	

Submission Deadline:

Hard Copy & Soft Copy: Sunday, May 31, 2026 till 11:59 PM

Instructions:

- **This assignment can be done in group.**
- **No late submission accepted.**
- **Viva can be conducted from all members of the group**
- **Attempt all questions in sequence. Attach this title page as a front page of assignment.**
- **Assignment should be handwritten on A4 sized page white paper.**
- **No pages from register please.**
- **Submit Hard copy in class and scanned copy on the portal.**
- **This title page should be attached.**

QNO:1

Part 1: Rules for number Codes ($\Sigma = \{0,1\}$).

(a)

Number must start with 1 and end 0.

R.E = $1.(0+1)^*.0$

(b)

Number must contain 11 at least once.

R.E = $(0+1)^*.11.(0+1)^*$

(c)

Number must contain an even number of 0s.

R.E = $1^*.(01^*.01^*)^*$

Part 2: Rules for player usernames ($\Sigma = \{a,b\}$).

(a)

Username cannot contain bb.

R.E = $(a+ba)^*.b?$

(b)

Username must end with a.

R.E = $(a+b)^*.a$

CamScanner

Question #02

Solution:-

a) DFA construction:

To satisfy both conditions simultaneously, the DFA must track whether a '1' has been read and the count of trailing zeros to prevent ending in '00'.

Formal Definition (5-tuple)

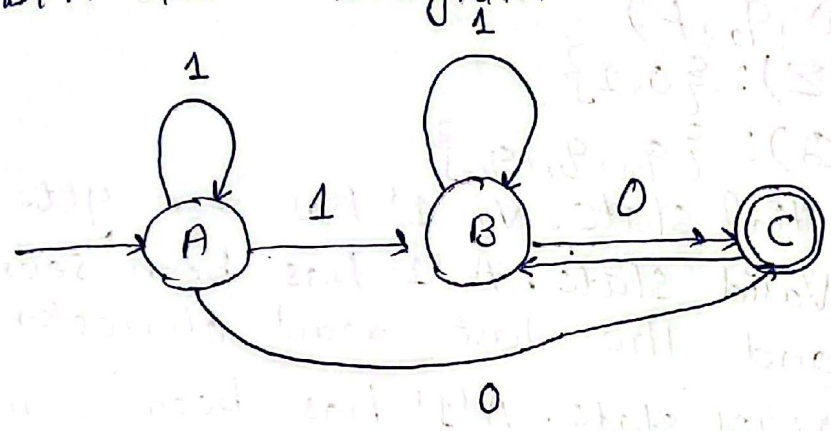
$M = (Q, \Sigma, \delta, q_0, F)$

- Alphabet (Σ): $\{0, 1\}$
- States (Q): $\{q_0, q_1, q_2\}$
  - q_0: Initial state. No '1' has seen yet.
  - q_1: Valid state. A '1' has been seen, and the last read character is '1'.
  - q_2: Valid state. A '1' has been seen, and the last read character is a single '0'.
  - q_3: Invalid state: A '1' hase been seen, but the string currently end in '00'.
- Initial State (q_0): q_0
- Final State (F): $\{q_1, q_2\}$

State Transition Table

Current state	Input 0	Input 1
→ q_0	q_0	q_1
* q_1	q_2	q_1
* q_2	q_3	q_1
q_3	q_3	q_1

DFA State Diagram



(b) Regular Expression Conversion

We drive the regular expression logically from the accepted paths in the DFA.

1. Every valid string must contain at least one '1'.
2. If a valid string ends with the character '1' it can be preceded by any combination of 0s and 1s. The regular expressions for this subset is $(0+1)^*1$.

3. If a valid string ends with the character '0', the character immediately preceding it must strictly be '1' to avoid forming '00' at the end. The regular expression for this subset is $(0+1)^* 10$.

Factoring out the common prefix the final minimal Regular Expression:

$$(0+1)^* 1 (\varepsilon + 0)$$

(c) DFA Trace

code 1: 101

- Start at q_0
- Read 1 → Transition to q_1
- Read 0 → Transition to q_2
- Read 1 → Transition to q_1
- Result: Ends in q_1. $q_1 \in F$. Accepted

Code 2: 100

- start at q_0
- Read 1 → Transition to q_1
- Read 0 → Trasition to q_2
- Read 1 → Transition to q_3

CamScanner

Result : Ends in $q_3 \cdot q_3 \notin F$. rejected.

Cod_3 : 0101

- Stat at q_0
- Read 0 → Transition to q_0
- Read 1 → Transition to q_1
- Read 0 → Transition to q_2
- Read 1 → Transition to q_1
- Result : Ends in $q_1 \cdot q_1 \in F$. Accepted.

Question 03

(a)

Moore machine design:-

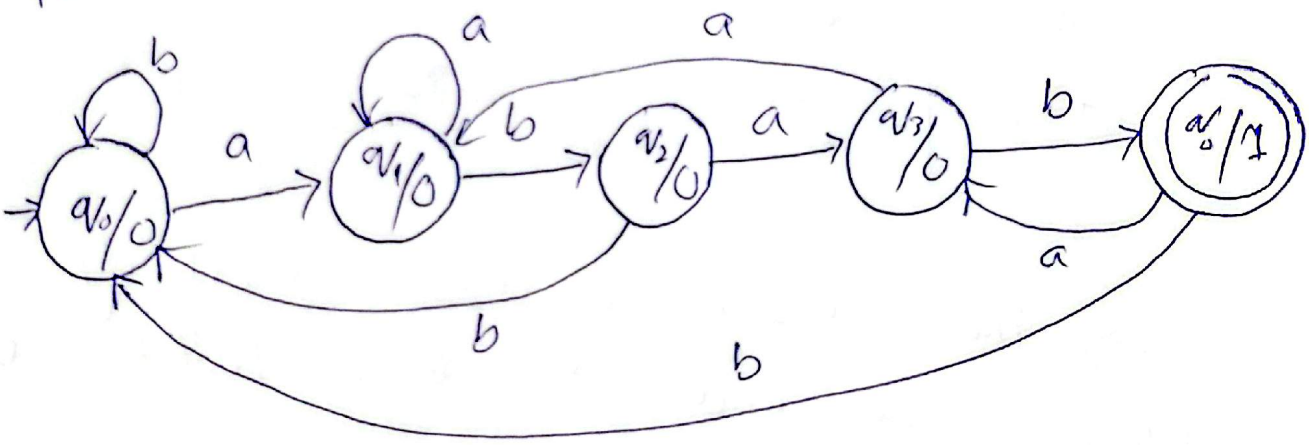
Pattern = "abab"

Where

a = SYN Packet

b = Ack Packet

Since this is a moore machine, the output depends on the state.



States:

State	output
q_0	0
q_1	0
q_2	0
q_3	0
q_4	1

CamScanner

(b)

Transition table:

Input accepted (Σ) : {a, b}

output Accepted (Δ) : {a, 1}

Present state	Next state		output
	input a	input b	
$\rightarrow q_0$	q_1	q_0	0
q_1	q_1	q_2	0
q_2	q_3	q_0	0
q_3	q_1	$+q_4$	0
$+q_4$	q_3	q_0	1

$\Rightarrow$ The moore machine successfully detects the malicious attack pattern as ab

Question # 04

Solution:

Given test Results:

Number (N)	Result
12	You win
8	Try again
15	Try again
24	Try again
22	you win

Analyzing Pattern:

- winning numbers: 12, 22
- Both are even numbers
- Both are greater than 10
- Not divisible by 5

Losing numbers: 8, 15, 24

- 8 → even but not > 10
- 15 → odd and divisible by 5
- 24 → even, > 10, but divisible by 5

CamScanner

IF (N % 2 == 0) AND (N > 10) AND (N % 5 != 0)

THEN You win!

Else

"Try again"

⇒ if <N % 2 == 0> AND <N > 10> AND <N % 5 != 0> → Then <You win>.

⇒ if WIN is True → You win!

if win is False → Try again

